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Evaluation of antibacterial and anticancer activity of green synthesized iron oxide (α -Fe₂O₃) nanoparticles

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ABSTRACT

In the present study, Piper betel leaves extract was used to synthesize iron oxide nanoparticle (α -Fe₂O₃) from anhydrous ferric chloride. The synthesized nanoparticles were characterized using XRD, SEM, TEM, FTIR and UV spectroscopy. The iron oxide nanoparticles obtained were cubical in shape and the XRD result showed the average size as 25.17 nm. The application potential of α -Fe₂O₃ nanoparticles prepared in the present study was tested for its antibacterial activity towards both gram negative (*Escherichia coli* and *Pseudomonas aeruginosa*) and gram positive bacteria (*Staphylococcus aureus* and *Streptococcus mutans*) by agar well diffusion method. Among the different bacterial pathogens used, the material exhibited highest antibacterial activity at a concentration of 1000 μ g/ml for *P. aeruginosa* (15 mm) while the lowest activity was observed for *Streptococcus mutans* (11 mm). The material was also investigated for its anticancer potential against A549 (Lung Cancer) cells by MTT assay. The observed cytotoxic response was in a concentration-dependent manner. The bioefficacy exhibited by the synthesized material may lead to its possible applications in environmental and biomedical fields.

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1. Introduction

Iron oxides are one of the most important transition metal oxides of biological importance. Among different polymorphs of iron oxide, hematite (α -Fe₂O₃) is the most stable one with tunable magnetic and optical properties, excellent chemical stability and biocompatibility that make it suitable for biomedical applications [1]. Several methods to synthesize iron oxide nanoparticles such as mechanochemical [2,3], wire electrical explosion [4], and electrochemical [5,6] have been reported; but these methods involve the use of toxic solvents and require high energy and vacuum conditions. Hence, there is a growing need to synthesize nanoparticles in a clean, non-toxic and ecofriendly way. A very wide range of biological resources like microorganisms (bacteria, yeast, fungi, algae, and viruses) and plants can be used for nanoparticle synthesis [7]. The low cost and easy availability of plants in nature make them more preferred biological resources than microbes [8]. The biomolecules involved with synthesis not only reduce the metal salt, but

also functionalize the surface of the nanoparticles, which induce synergistic effects for antimicrobial or cancer treatment applications [9]. Our previous studies also focused on plant mediated preparation of nanoparticles [10]. Though there are various reports on plant mediated synthesis of iron oxide nanoparticles, only a few give emphasis on anticancer and antibacterial activity of hematite nanoparticles.

In the present study, Piper betel leaves extract was used to synthesize α -Fe₂O₃ nanoparticle from anhydrous ferric chloride. The biomolecules present in the leaf extract are responsible for the pharmacological, laxative and anti-helminthic activities [10]. The phenolic compounds present in the leaf extract act as reducing as well as capping agent facilitating the formation of nanoparticles.

Several metal oxide nanoparticles have been known to possess antimicrobial properties. They have the capability to overcome bacterial drug resistance due to their multi-functionality. The effectiveness on resistant strains of microbial pathogens, robustness and stability of inorganic oxides nanoparticles make them more preferential to organic antimicrobial agents [11]. In this regard, iron oxide nanoparticles obtained in the present study were tested for antibacterial activity.

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Cancer is now the second leading cause of human death worldwide [12]. Nanomedicine plays an important role in developing alternative and more effective treatment strategies for cancer [13]. Green synthesized nanoparticles are used in cancer treatment since they are eco-friendly, biocompatible and of low cost. Among the various nano metal oxides, iron oxide nanoparticles have been found to be promising due to their wide range of application as antioxidant, antibiofilm, antibacterial, and anticancer agents [14–18]. In this background, the present study aims to evaluate the antibacterial and anticancer activity of hematite nanoparticles synthesized via green method. The anti cancer activity of hematite nanoparticles was examined in A549 lung cancer cells by MTT {3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyl tetrazolium bromide} assay.

2. Materials and methods

2.1. Materials

The only chemical used for the synthesis of hematite nanoparticles was anhydrous ferric chloride [FeCl_3] of analytical grade. Bacterial culture media were purchased from HiMedia (Mumbai, India). Dimethyl sulphoxide (DMSO), antibiotic solution and Dulbecco's Modified Eagle's Medium (DMEM) were purchased from Sigma aldrich, USA. Fetal Bovine Serum was obtained from Gibco, US origin and Trypsin from Invitrogen, USA 25200-056. MTT was purchased from Sigma Aldrich M-5655. Culture plates and flasks were obtained from Nunc, ThermoScientific USA. A549 (Lung Cancer) cells were procured from National Centre for Cell Sciences (NCCS), Pune, India.

2.2. Green synthesis of hematite nanoparticles

Piper betel leaf extract was prepared with slight modifications to the method suggested in the previous report [10]. Piper betel leaves were collected, washed several times with pure water to remove dust particles, dried in shade and then pulverised. 110 mL of distilled water was added to 10 g of powdered leaves and heated in a microwave oven at 100 W for 30 min. The mixture was filtered using a Whatman 1 filter paper and the filtrate was collected. This freshly prepared leaf extract was then used for the synthesis of hematite nanoparticles.

Freshly prepared piper betel leaf extract was added to 50 mL 0.1 M aqueous solution of Ferric Chloride with stirring using a magnetic stirrer for 1 h. The orange coloured solution of ferric chloride changed to brownish black on the addition of leaf extract, with the formation of a black coloured precipitate. The mixture was then sonicated for 30 min, filtered, washed with distilled water and dried. The dried precipitate was heated in a muffle furnace at 600 °C for 3 h to obtain iron oxide nanoparticles.

The possible mechanism of formation of $\alpha\text{-Fe}_2\text{O}_3$ nanoparticles is as follows. Initially, octahedral aqua complex of Fe(III) , $\text{Fe}(\text{H}_2\text{O})_6$ is formed instantaneously in water [1]. The hydrolysed iron species could form a complex with the functional groups present in Piper betel leaf extract. With the increase in temperature, a phase transformation of iron species into $\text{Fe}(\text{OH})_2$ might occur which eventually form primary particles (Fe_2O_3) with nano scale dimension.

2.3. Biophysical characterization of $\alpha\text{-Fe}_2\text{O}_3$ nanoparticles

The synthesized nanoparticles were characterized by XRD, SEM, TEM, SAED, FTIR and UV spectroscopy. XRD was recorded using Bruker AXS D8 Advance model diffractometer. Morphology and particle size of the prepared material was examined using SEM of make JEOL Model JSM-6390LV and TEM of make Jeol/JEM 2100.

Selected area diffraction (SAED) pattern was also obtained. The involvement of organic functional groups in the formation of nanomaterials was analysed by FTIR spectrum which was recorded using Thermo Nicolet, Avatar 370 in the spectral range of 4000–400 cm^{-1} . The optical properties of the sample were evaluated using UV Visible spectrophotometer of make Varian Cary 500.

2.4. Antibacterial activity

Antibacterial activity of the synthesized iron oxide nanoparticles was assessed using agar well diffusion method. Pathogens namely *E.coli* [ATCC 25922], *Pseudomonas aeruginosa* [ATCC 27853] (gram negative bacteria) and *Streptococcus mutans* [MTCC 890] and *Staphylococcus aureus* [ATCC 25923] (gram positive bacteria) were seeded on petriplates containing Muller Hinton Agar Medium (growth of culture adjusted according to McFarland Standard, 0.5%). Agar surface was bored using a well cutter to make wells of approximately 10 mm diameter and different concentrations of sample such as 250 $\mu\text{g/mL}$, 500 $\mu\text{g/mL}$ and 1000 $\mu\text{g/mL}$ were added. The plates were then incubated at 37 °C for 24 h and were examined for antibacterial activity by measuring the diameter of the inhibition zone (expressed in millimeter) formed around the well. Streptomycin was used as the positive control.

2.5. Anticancer activity

A549 (lung cancer) cells were used to determine the cell viability against the exposure of hematite nanoparticles. The cells were maintained in Dulbecco's modified Eagles medium, DMEM. The cell lines were cultured with DMEM supplemented with 10% fetal bovine serum (FBS), sodium bicarbonate, L-glutamine, and antibiotic solution containing: Streptomycin (100 mg/mL), Penicillin (100 U/mL), and Amphotericin B (2.5 mg/mL). The cultured cell lines were kept at 37 °C in a humidified 5% CO_2 incubator. Briefly, 5000 A549 cells per well were seeded in 96 well plate and incubated at 37 °C for 48 h in a humidified 5% CO_2 incubator.

After seeding, the monolayer of the cells in the plate was treated with different concentrations (6.25, 12.5, 25, 50 and 100 $\mu\text{g/mL}$) of hematite nanoparticles and the plate was incubated for 24 h at 37 °C in CO_2 incubator. Non treated control cells were also maintained. MTT {3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyl tetrazolium bromide} reagent was used to evaluate cell viability [19]. After incubation, the sample content in wells was discarded and 30 μL of MTT (15 mg/mL in Phosphate buffered saline) was added to each well and incubated at 37 °C in a humidified 5% CO_2 incubator for 4 h. After the incubation period, the supernatant was removed and 100 μL of DMSO was added to each well to dissolve the obtained formazan crystals. The optical density values were measured by using microplate reader at a wavelength of 540 nm. The relative cell viability was calculated as the absorbance ratio of test sample to that of the control [20].

The viability of cells was also evaluated by direct observation of cells by Inverted phase contrast microscope (Olympus CKX41 with Optika Pro5 CCD camera) and microscopic observations were recorded as images. Any detectable change in the morphology of the cells, such as granulation, shrinking or rounding of cells, and vacuolization in the cytoplasm of the cells were considered as indicators of cytotoxicity.

3. Results and discussion

3.1. XRD of $\alpha\text{-Fe}_2\text{O}_3$ nanoparticles

The XRD pattern of $\alpha\text{-Fe}_2\text{O}_3$ (hematite) nanoparticles is shown in Fig. 1. The XRD peaks at 24.2°, 33.18°, 35.67°, 40.81°, 49.47°,

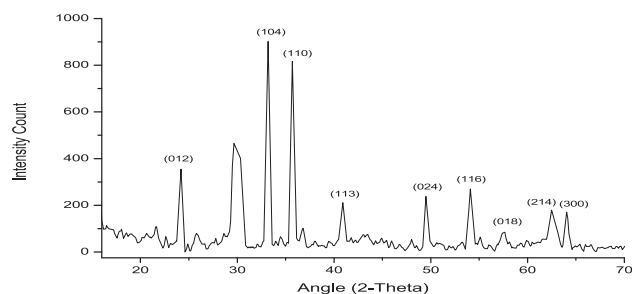


Fig. 1. XRD of nano α -Fe₂O₃.

54.1°, 57.60°, 62.46°, and 64.13° 2 θ are assigned to the (0 1 2), (1 0 4), (1 1 0), (1 1 3), (0 2 4), (1 1 6), (0 1 8), (2 1 4) and (3 0 0) planes, respectively (Fig. 1). These results indicate a rhombohedral α -Fe₂O₃ phase and match with JCPDS card no. 33-0664. The mean size of hematite nanoparticles estimated using Debye-Scherrer formula was found to be 25.17 nm. Similar XRD results have also been reported earlier [20].

3.2. SEM of α -Fe₂O₃ nanoparticles

The SEM image (Fig. 2) shows almost cubic morphology for the prepared nano α -Fe₂O₃.

3.3. TEM of α -Fe₂O₃ nanoparticles

The TEM image indicates that the prepared nanoparticles are cubic in shape as shown in Fig. 3 with a particle size of 22 to 35 nm. This result is in good agreement with the particle size calculated using Debye-Scherrer's equation. Most of the α -Fe₂O₃ nanoparticles were found to be cubical with smooth edges. The SAED pattern in Fig. 4 confirms the nanocrystalline nature of the synthesized material.

3.4. FTIR spectrum of nano α -Fe₂O₃

The FTIR spectrum of nano Fe₂O₃ is shown in Fig. 5. The infrared spectrum (FTIR) of the synthesized Fe₂O₃ nanoparticles recorded in the wavenumber range of 400–4000 cm⁻¹ identifies the chemical bonds as well as functional groups in the compound. The broad band at 3437 cm⁻¹ is assigned to O–H stretching vibration indicating the presence of absorbed water molecule. The peaks at 1635 cm⁻¹ and 1091 cm⁻¹ can be attributed to C=C and C–N stretching of bioactive molecules present on the surface of the nanoparticles [21]. The two absorption bands below 700 cm⁻¹ represent the characteristic features of α -Fe₂O₃ and are assigned to metal oxygen stretching frequencies [22]. The high frequency band ~ 560 cm⁻¹ refers to Fe–O deformation in the octahedral

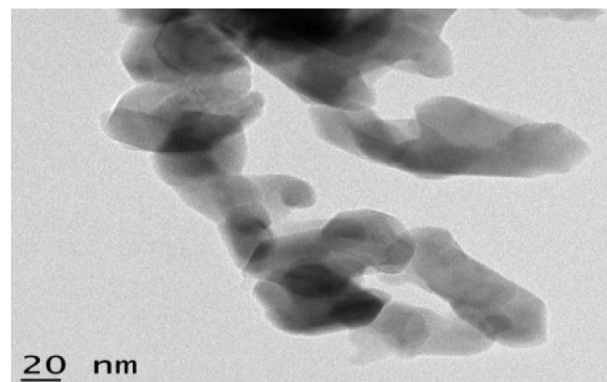


Fig. 3. TEM image of nano α -Fe₂O₃.

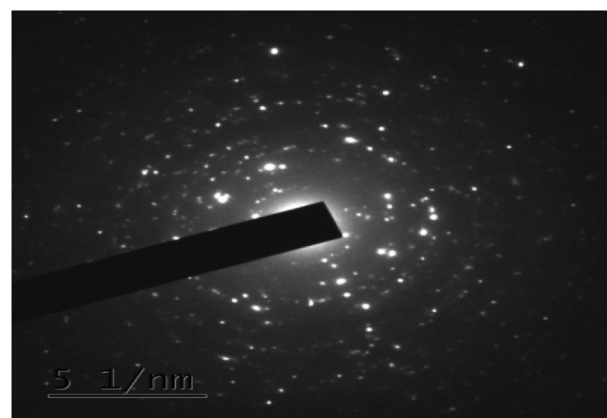


Fig. 4. SAED pattern of nano α -Fe₂O₃.

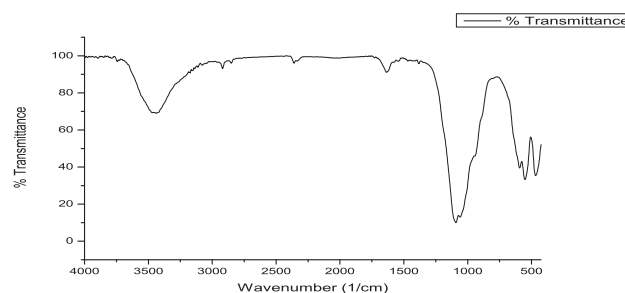


Fig. 5. FTIR spectrum of nano α -Fe₂O₃.

and tetrahedral sites while the low frequency band ~ 462 cm⁻¹ is attributed to Fe–O deformation in the octahedral site of hematite [23].

3.5. UV-visible diffuse reflectance spectroscopy

The UV-visible diffuse reflectance spectrum of nano Fe₂O₃ is shown in Fig. 6. The strong absorption peak in the wavelength of ~ 400 nm corresponds to α -Fe₂O₃ nanoparticles [22]. Several factors including particle size, shape etc. are responsible for the broadening of the peak [20].

The band gap energy of the sample was estimated by extrapolating the linear region at the steeply increasing curve of Tauc's plot as shown in Fig. 7. The value of the gap is given by the X-intercept and is found to be 2.3 eV. This value is slightly higher than that of bulk hematite (2.2 eV) [24]. The observed blue shift is due to quantum confinement effect [25].

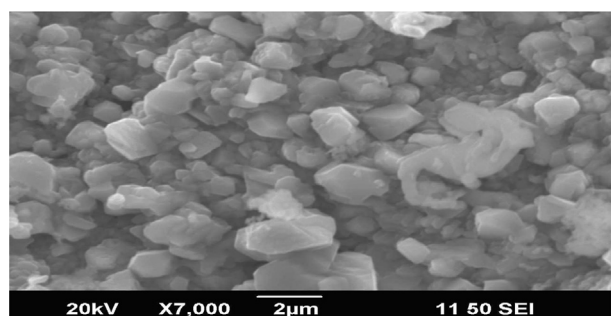


Fig. 2. SEM image of nano α -Fe₂O₃.

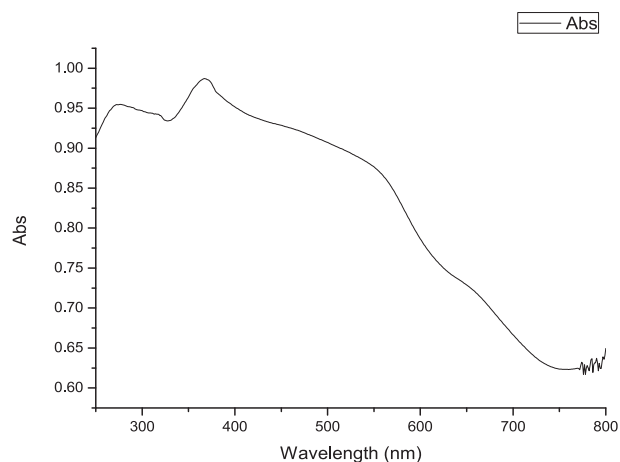


Fig. 6. UV-visible diffuse reflectance spectrum of nano α -Fe₂O₃.

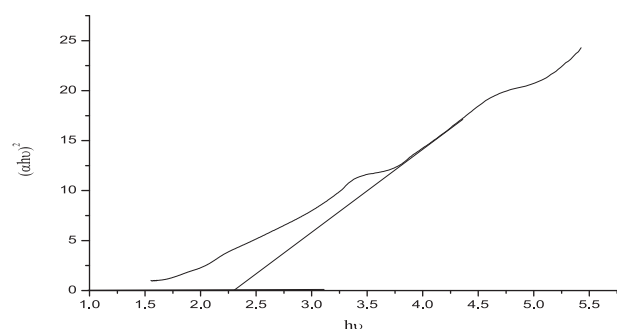


Fig. 7. Tauc's plot of nano α -Fe₂O₃.

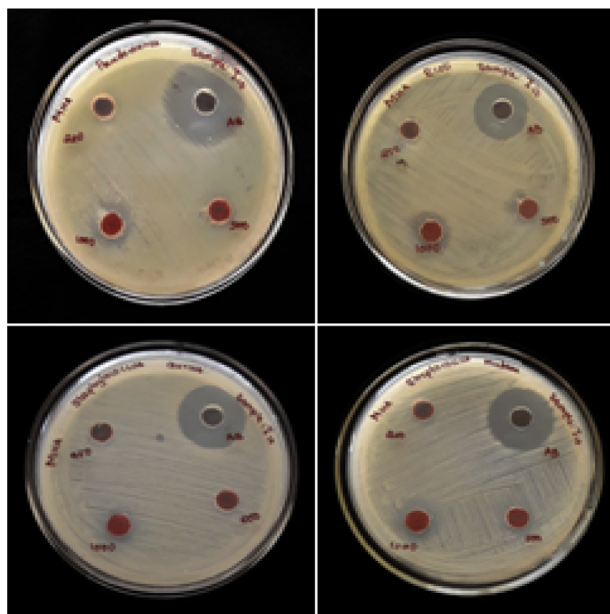


Fig. 8. Zone inhibition image of *P. aeruginosa*, *E. coli*, *S. aureus* and *S. mutans* in the presence of nano α -Fe₂O₃.

3.6. Antibacterial activity

The obtained material has been examined for its antibacterial activity against *E. coli*, *P. aeruginosa*, *S. aureus* and *S. mutans* using agar well-diffusion method and significant zone of inhibitions

Table 1

Tabulation showing zone of inhibition of *E. coli*, *P. aeruginosa*, *S. aureus* and *S. mutans* in the presence of nano Fe₂O₃.

Pathogens	Zone of inhibition (mm)			
	Concentration (µg/ mL)			Control
	250	500	1000	
Escherichia coli	–	–	13	25
Pseudomonas aeruginosa	–	12	15	31
Staphylococcus aureus	–	–	12	27
Streptococcus mutans	–	–	11	28

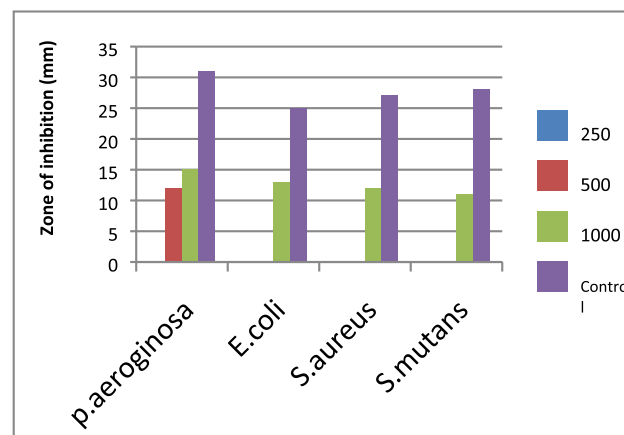


Fig. 9. Effect of nano α -Fe₂O₃ on the growth of different bacterial strains.

(ZOI) were observed as shown in Fig. 8. The reduction of bacterial growth rate on increasing the concentration of iron oxide nanoparticles is evident from Table 1.

The zone of inhibition of bacterial growth was not observed at lower concentrations of α -Fe₂O₃ except for *P. aeruginosa* for which a ZOI of 12 mm was obtained at 500 μ g/mL. Among the different bacterial pathogens used, α -Fe₂O₃ NPs was more efficient to inhibit the growth of *P. aeruginosa* (15 mm) while the lowest activity was observed for *S. mutans* (11 mm) as evident from the graph (Fig. 9).

There are two possible mechanisms for the interaction between hematite (α -Fe₂O₃) NPs and bacteria. One among them is the production of increased levels of reactive oxygen species (ROS) including hydroxyl radicals (OH[•]), hydrogen peroxide (H₂O₂) and singlet oxygen (O₂[•]). When hematite nanoparticles with defects are activated by UV or visible light, electron-hole pairs are created which in turn contribute to the generation of reactive oxygen species. The generated free radicals can depolymerize polysaccharides, inactivate enzymes, cause DNA strands break, and kick off lipid peroxidation leading to bacterial death [26]. The other feasible mechanism involves electrostatic, dipole-dipole, hydrogen bond, hydrophobic and vander waal's interactions which can cause disruption of cellular function and disorganization of membranes [1].

3.7. Anticancer activity

The cytotoxicity of hematite nanoparticles was evaluated against human A-549 cancer cells at various concentrations (1–100 μ g/mL). The cell proliferation was examined using an inverted phase contrast microscope. Microscope photographs of A549 cells after treatment with nano α -Fe₂O₃ (Fig. 10) showed cell shrinkage, condensed and fragmented nuclei.

Significant morphological changes were not observed at lower concentrations (1–10 μ g/mL) of hematite. The material caused

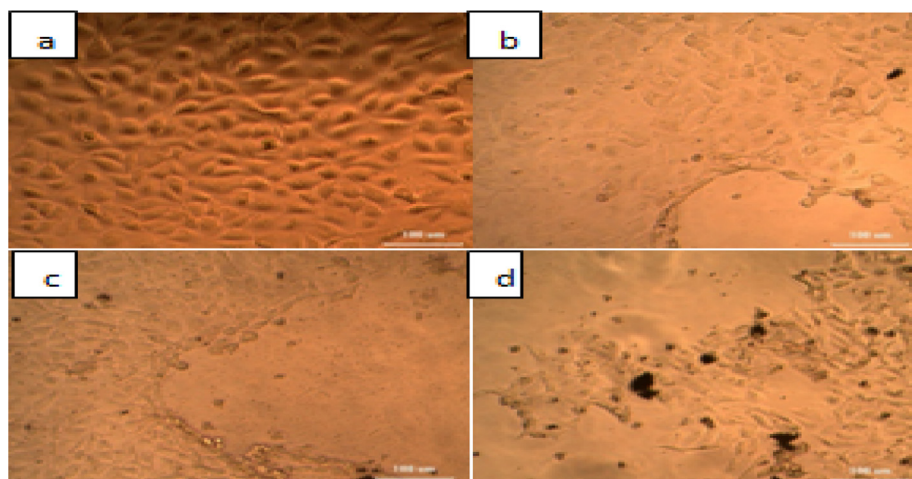


Fig. 10. Images of cells (a) without Fe_2O_3 NP and at various concentrations b) 25 $\mu\text{g/mL}$ c) 50 $\mu\text{g/mL}$ d) 100 $\mu\text{g/mL}$.

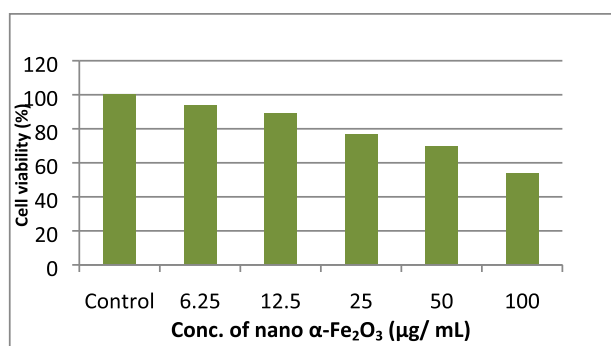


Fig. 11. Effect of nano Fe_2O_3 on viability of A549 cells determined by MTT assay.

cytotoxicity in A549 lung cancer cells in a concentration dependent manner as evident from Fig. 11. $\alpha\text{-Fe}_2\text{O}_3$ nanoparticles showed 50% of cell death at 104.6 $\mu\text{g/mL}$ concentration, which is considered as the IC_{50} value. The synthesized nanoparticles were found to be moderately active as per the activity levels labeled by US National Centre Institute [27].

The size and dosage of nanoparticle play a key role in inducing toxicity [28,29]. Clathrin-mediated transport, macro-pinocytotic uptake and caveolae-dependent endocytosis are some of the ways in which the nanoparticles gain its entry into the cells [30,31]. Once inside the cell, hematite nanoparticles can directly interact with NADPH oxidases from the plasma membrane and/or mitochondria and result in the generation of superoxide anion [32,33]. This, in turn, activates redox-sensitive signaling cascades and leads to pro-inflammatory effects and cytotoxicity [34,35].

4. Conclusion

The focus of the present work is on an efficient and cost effective green synthesis of hematite ($\alpha\text{-Fe}_2\text{O}_3$) nanoparticles and investigation of its antibacterial and anticancer activity. XRD and SEM studies confirmed the formation of nanoparticles of cubical shape with crystallite size of 25.17 nm. The antibacterial efficacy of the nanoparticles against Gram-positive as well as Gram negative bacteria is also established. The synthesized hematite nanoparticles exerted cytotoxic effect on A549 lung cancer cells in a concentration dependent manner. The bioefficacy exhibited by the synthe-

sized $\alpha\text{-Fe}_2\text{O}_3$ nanoparticles may lead to its possible applications in environmental and biomedical fields.

CRediT authorship contribution statement

Jumna Yoonus: Conceptualization, Methodology, Writing - original draft. **R. Resmi:** Writing - review & editing. **B. Beena:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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